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Morphological Analysis for Indian Languages

Technical presentation

DSA0302 - NATURAL LANGUAGE PROCESSING

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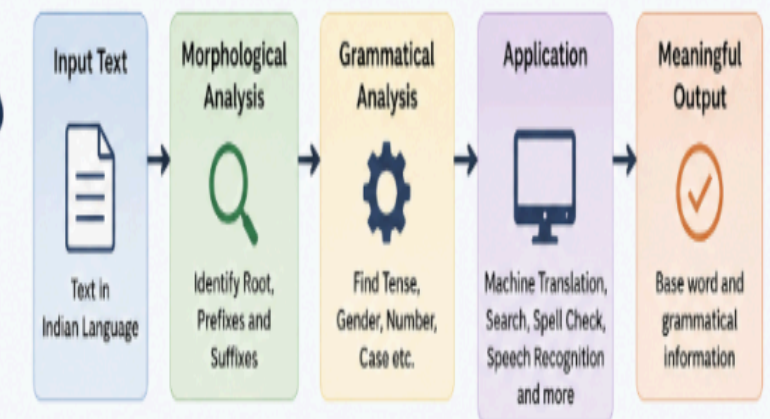
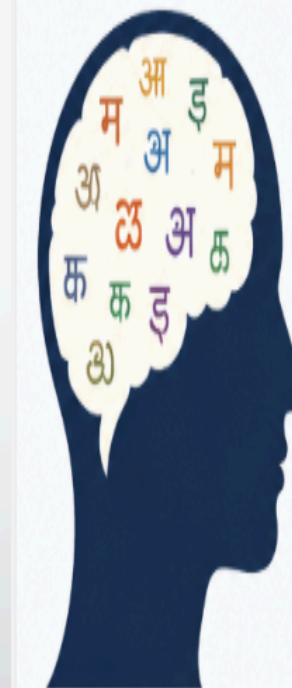
INTRODUCTION



- ▶ Morphological Analysis is the process of identifying the root word, prefixes, and suffixes in a word.
- ▶ It helps computers understand the structure and meaning of words.
- ▶ Indian languages such as Tamil, Telugu, Hindi, Kannada, and Malayalam are morphologically rich, with many word variations.
- ▶ A single root word can have multiple forms based on tense, gender, number, and case.
- ▶ Morphological analysis improves the accuracy of Natural Language Processing (NLP) applications.

Indian Languages

Morphological Analysis is the process of breaking words into their smallest meaningful units (morphemes) to understand their structure and meaning.



Examples	Hindi	Tamil	Telugu	Kannada
	लड़कौं	புத்தகங்களில்	పిల్లలకు	ಮನೆಗಲ್ಲ
	लड़का + औं	புத்தகம் + கள் + இல்	పిల్ల + ల + కు	ಮನೆ + ಗಲ್ಲ
	Root Suffix (Plural)	Root Suffix (Plural) Suffix (In)	Root Suffix (Plural) Suffix (To)	Root Suffix (In)



Problem Statement

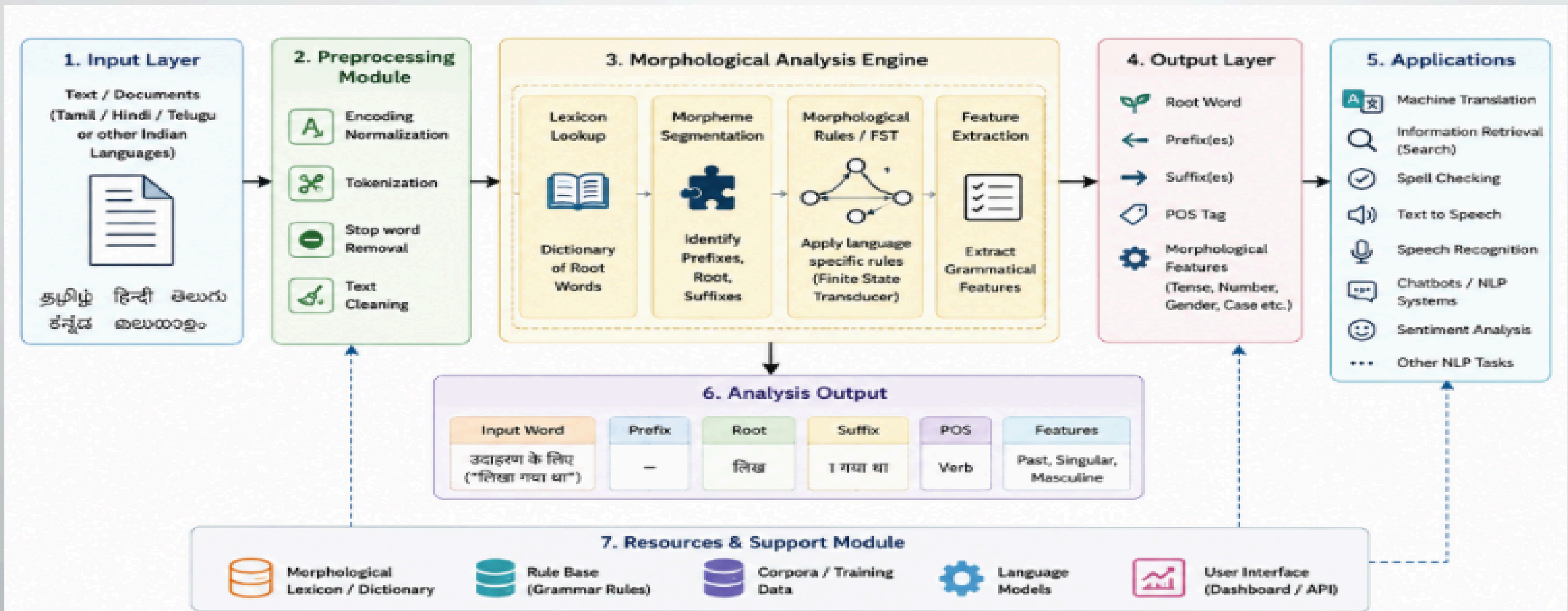
- ▶ Indian languages have complex word structures with many prefixes, suffixes, and inflections.
- ▶ A single root word can appear in multiple grammatical forms, making automatic text processing difficult.
- ▶ Traditional NLP techniques often fail to accurately identify the root word and its grammatical information.
- ▶ Differences in grammar and vocabulary across Indian languages increase the complexity of language processing.
- ▶ Limited availability of language resources and annotated datasets affects the performance of NLP systems.



Objectives

- ▶ To understand the concept of morphological analysis in Natural Language Processing (NLP).
- ▶ To identify the root words, prefixes, and suffixes in Indian language words.
- ▶ To analyze the grammatical structure of words in languages such as Tamil, Telugu, and Hindi.
- ▶ To improve the accuracy of text processing by identifying different word forms.
- ▶ To support NLP applications such as machine translation, spell checking, search engines, and speech recognition.

System Architecture





Result

- ▶ Successfully identified the root words, prefixes, and suffixes from Indian language words.
- ▶ Improved understanding of morphologically rich languages such as Tamil, Telugu, and Hindi.
- ▶ Correctly extracted grammatical features like tense, number, gender, and case.
- ▶ Reduced ambiguity in word interpretation through morphological parsing.
- ▶ Enhanced the performance of machine translation and information retrieval systems.
- ▶ Improved the accuracy of spell checking, speech recognition, and text processing applications.



Conclusion

- ▶ Morphological analysis is an essential component of Natural Language Processing (NLP) for Indian languages.
- ▶ It helps identify the root words, prefixes, and suffixes, improving language understanding.
- ▶ It effectively handles the complex grammatical structures of Tamil, Telugu, Hindi, and other Indian languages.
- ▶ Morphological analysis improves the accuracy of applications such as machine translation, spell checking, search engines, and speech recognition.
- ▶ It supports the development of AI-based multilingual systems for regional languages.



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THANK YOU